Dr. Taylor Smith

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ORCID: 0000-0002-6763-7204

Research Profile

My ongoing research is focused on monitoring global vegetation systems, quantifying the resilience of natural ecosystems to climate change, and understanding the hydrology of alpine regions. My methods are data-driven and rely on the combination of high-performance computing and novel statistical methods to explore changes in the Earth system.

Education

2018 **Dr. rer. nat. (*summa cum laude*), Remote Sensing**, *Universität Potsdam*, Germany
2014 **Master of Arts, Geography**, *UC Santa Barbara*, USA
2011 **Bachelor of Arts, Geology**, *Middlebury College*, USA

Professional Experience

2026 – **Research Staff**, *Technical University of Munich*, Germany
2018 – **Research Staff and Lecturer**, *Universität Potsdam*, Germany
2014–2018 **PhD Candidate and Teaching Assistant**, *Universität Potsdam*, Germany
2013–2018 **Independent Consultant, Climate Hazard Analysis for International Development**, *Remote*
Summer 2013 **Natural Hazards Research Fellow**, *Earth Research Institute*, USA
2012–2014 **MA Candidate and Teaching Assistant**, *UC Santa Barbara*, USA
2011–2012 **GIS Analyst**, *Tetra Tech ARD*, USA
Summer 2010 **Field Researcher**, *UMass Amherst*, USA
2009–2011 **Research and Teaching Assistant**, *Middlebury College*, USA

Funding

2023–2026 **DFG**: Principle Investigator, "Quantifying the Influence of Snowmelt on River Hydrology in High Mountain Asia" (EUR 365k)
2019–2022 **BMBF**: Postdoctoral Fellow, "Options for Sustainable Land Use Adaptations in Savanna Systems" (ORYCS)
2013 **ERI**: Natural Hazards Research Fellow, "Mapping Glacial Lake Outburst Floods in the Himalaya" (USD 5k)

Awards, Honors, and Certifications

2021 **International Teaching Professional** (U. Potsdam) – Advanced training for University Pedagogy.
2018 **Michelson Prize** (U. Potsdam) – Awarded for the top doctoral thesis of the year in the Math and Science Faculty at Universität Potsdam.
2018 **Prize for Excellence in Teaching** (U. Potsdam) – Awarded for excellent teaching at the graduate level. Assessed based on course design, innovation, and student reviews.

Peer-Reviewed Publications

Journal Articles

2026 [33] **T Smith**, A Morr, C Schötz, and N Boers. "Estimating the Resilience of Non-Stationary Systems." *Nature Communications* 17, 11286 (2026). https://doi.org/10.1038/s41467-026-78489-7

[32] Z Guo, J Wang, **T Smith**, W Li, K Zhang, J Wu, Y Zhou, and P Gong "Widespread reversal of vegetation resilience in permafrost ecosystems." *Nature Plants* (2026). https://doi.org/10.1038/s41477-026-02462-x

[31] **T Smith** and B Bookhagen. "Discharge deficits in the Central Himalaya linked to greening and warming." *Scientific Reports* (2026). https://doi.org/10.1038/s41598-026-68997-3

[30] S Ma, S Zhou, D Ellison, A Staal, **T Smith**, C Smith, and A J Teuling. "Targeted tropical forest restoration can offset deforestation-induced water flux losses." *Nature Climate Change* (2026). https://doi.org/10.1038/s41558-026-02709-7

[29] S Svintsov, P Pradhan, **T Smith**, and D Rybski. "Integrating agriculture into European urban landscapes matters: A systematic assessment." *Sustainable Cities and Society* (2026). https://doi.org/10.1016/j.scs.2026.107422

[28] J Wang, C Zhang, Y Pan, Y Su, P Ciais, **T Smith**, J Shang, J Liu, J M Chen, A Cescatti, Y Zhang, W F Laurance, C Senf, Z Zuo, J Liao, R Lafortezza, K Yan, P Zhu, X Guan, X Li, M Xue, W Yuan, X Chen, and W Zhou. "The safety margin of small-scale tree cover loss in global forests." *Nature Communications* (2026). https://doi.org/10.1038/s41467-026-71480-2

[27] T Liu, A Morr, S Bathiany, L Blaschke, Z Qian, C Diao, **T Smith**, and N Boers. "Data gaps and outliers distort critical slowing down-based resilience indicators." *Science Advances* 12, eaee1916 (2026). https://doi.org/10.1126/sciadv.aee1916

[26] **T Smith**, A Morr, B Bookhagen and N Boers. "Predicting Instabilities in Transient Landforms and Interconnected Ecosystems." *Nature Communications* 17, 1316 (2026). https://doi.org/10.1038/s41467-026-68944-w

2025 [25] **T Smith** and B Bookhagen. "Strongly Heterogeneous Surface-Water Warming Trends in High Mountain Asia." *Geophysical Research Letters* (2025). https://doi.org/10.1029/2025GL119418

[24] N Boers, T Liu, S Bathiany, M Ben-Yami, L Blaschke, N Bochow, C Boulton, T Lenton, A Morr, D Nian, M Rypdal, and **T Smith**. "Destabilization of Earth system tipping elements." *Nature Geoscience* (2025). https://doi.org/10.1038/s41561-025-01787-0

[23] L Flores, C Nendel, B Bookhagen, J A Oviedo-Reyes, **T Smith**, and G Ghazaryan. "The potential of Sentinel-1 time series for large-scale assessment of maize and wheat phenology across Germany." *GIScience & Remote Sensing*, 62(1) (2025). https://doi.org/10.1080/15481603.2025.2531593

[22] Y Su, C Zhang, A Cescatti, K Yu, P Ciais, **T Smith**, J Shang, J Carnicer, J Liu, JM Chen, J K Green, J Wu, G E Ponce-Campos, Y Zhang, Z Zuo, J Liao, J Wu, R Lafortezza, K Yan, X Yang, L Liu, J Ren, W Yuan, X Chen, C Wu, and W Zhou. "Pervasive but biome-dependent relationship between fragmentation and resilience in forests." *Nature Ecology & Evolution* (2025). https://doi.org/10.1038/s41559-025-02776-7

2024 [21] L Blaschke, D Nian, S Bathiany, M Ben-Yami, **T Smith**, C Boulton, and N Boers. "Spatial correlation increase in single-sensor satellite data reveals loss of Amazon rainforest resilience." *Earth's Future*, 12, e2023EF004040 (2024). https://doi.org/10.1029/2023EF004040

[20] S Bathiany, R Bastiaansen, A Bastos, L Blaschke, J Lever, S Loriani, W De Keersmaecker, W Dorigo, M Milenković, C Senf, **T Smith**, J Verbesselt, and N Boers. "Ecosystem Resilience Monitoring and Early Warning Using Earth Observation Data: Challenges and Outlook." *Surveys in Geophysics* (2024). https://doi.org/10.1007/s10712-024-09833-z

[19] T M Lenton, J F Abrams, A Bartsch, S Bathiany, C A Boulton, J E Buxton, A Conversi, A M Cunliffe, S Hebden, T Lavergne, B Poulter, A Shepherd, **T Smith**, D Swingedouw, R Winkelmann, and N Boers. "Remotely sensing potential climate change tipping points across scales." *Nature Communications* 15, 343 (2024). https://doi.org/10.1038/s41467-023-44609-w

2023 [18] Y Yang, Q You, **T Smith**, R Kelly, S Kang. "Spatiotemporal dipole variations of spring snowmelt over Eurasia." *Atmospheric Research*: 107042 (2023). https://doi.org/10.1016/j.atmosres.2023.107042

[17] **T Smith** and N Boers. "Reliability of Vegetation Resilience Estimates Depends on Biomass Density." *Nature Ecology & Evolution*: 1-10 (2023). https://doi.org/10.1038/s41559-023-02194-7

[16] T Schmidt, T Kuester, **T Smith**, and M Bochow. "Potential of Optical Spaceborne Sensors for the Differentiation of Plastics in the Environment." *Remote Sensing* 15(8):2020 (2023). https://doi.org/10.3390/rs15082020

[15] **T Smith**, R.-M. Zotta, C. A. Boulton, T. M. Lenton, W. Dorigo, and N. Boers. "Reliability of resilience estimation based on multi-instrument time series." *Earth System Dynamics*, 14, 173–183 (2023). https://doi.org/10.5194/esd-14-173-2023

[14] **T Smith** and N Boers. "Global vegetation resilience linked to water availability and variability." *Nature Communications* 14, 498 (2023). https://doi.org/10.1038/s41467-023-36207-7

2022 [13] **T Smith**, D Traxl, and N Boers. "Empirical evidence for recent global shifts in vegetation resilience." *Nature Climate Change* (2022). https://doi.org/10.1038/s41558-022-01352-2

[12] R Hering, M Hauptfleisch, M Jago, **T Smith**, S Kramer-Schadt, J Stiegler, and N Blaum. "Don't stop me now: Managed fence gaps could allow migratory ungulates to track dynamic resources and reduce fence related energy loss." *Frontiers in Ecology and Evolution* (2022). https://doi.org/10.3389/fevo.2022.907079

[11] F Atmani, B Bookhagen, **T Smith**. "Measuring Vegetation Heights and Their Seasonal Changes in the Western Namibian Savanna Using Spaceborne Lidars." *Remote Sensing* 14, 2928 (2022). https://doi.org/10.3390/rs14122928

2021 [10] **T Smith**, A Rheinwalt, and B Bookhagen. "Topography and Climate in the Upper Indus Basin: Mapping Elevation-Snow Cover Relationships." *Science of The Total Environment*, 147363 (2021). https://doi.org/10.1016/j.scitotenv.2021.147363

2020 [9] **T Smith** and B. Bookhagen. "Climatic and Biotic Controls on Topographic Asymmetry at the Global Scale." *Journal of Geophysical Research: Earth Surface*, 125, e2020JF005692 (2020). https://doi.org/10.1029/2020JF005692

[8] **T Smith** and B. Bookhagen. "Assessing Multi-Temporal Snow-Volume Trends in High Mountain Asia From 1987 to 2016 Using High-Resolution Passive Microwave Data." *Front. Earth Sci.* 8:559175 (2020). https://doi.org/10.3389/feart.2020.559175

2019 [7] **T Smith**, A Rheinwalt, and B Bookhagen. "Determining the Optimal Grid Resolution for Topographic Analysis on an Airborne Lidar Dataset." *Earth Surface Dynamics* 7: 475-489 (2019). https://doi.org/10.5194/esurf-7-475-2019

2018 [6] **T Smith** and B Bookhagen. "Using passive microwave data to understand spatio-temporal trends and dynamics in snow-water storage in High Mountain Asia." *Proc. SPIE 10788*, 1078806 (2018). https://doi.org/10.1117/12.2323827

[5] **T Smith** and B Bookhagen. "Changes in seasonal snow water equivalent distribution in High Mountain Asia (1987 to 2009)." *Science Advances* 4: 1 (2018). https://doi.org/10.1126/sciadv.1701550

2017 [4] **T Smith**, B Bookhagen, and A Rheinwalt. "Spatio-temporal Patterns of High Mountain Asia's Snowmelt Season Identified with an Automated Snowmelt Detection Algorithm, 1987-2016." *The Cryosphere* 11: 2329-2343 (2017). https://doi.org/10.5194/tc-11-2329-2017

2016 [3] **T Smith** and B Bookhagen. "Assessing uncertainty and sensor biases in passive microwave data across High Mountain Asia." *Remote Sensing of Environment* 181: 174-185 (2016). https://doi.org/10.1016/j.rse.2016.03.037

2015 [2] **T Smith**, B Bookhagen, and F Cannon. "Improving semi-automated glacier mapping with a multi-method approach: applications in central Asia." *The Cryosphere* 9.5: 1747-1759 (2015). https://doi.org/10.5194/tc-9-1747-2015

2013 [1] W Amidon, B Bookhagen, J-P Avouac, **T Smith**, D Rood. "Late Pleistocene Glacial Advances in the Western Tibet Interior." *Earth and Planetary Science Letters* 381: 210-221 (2013). https://doi.org/10.1016/j.epsl.2013.08.041

Book Chapters

2024 [2] K Geißler, N Blaum, G von Maltitz, **T Smith**, B Bookhagen, H Wanke, M Hipondoka, E Hamunyelae, D Lohmann, D U Lüdtke, M Mbidzo, M Rauchecker, R Hering, K Irob, B Tietjen, A Marquart, F V Skhosana, T Herkenrath and S Uugulu. "Biodiversity and Ecosystem Functions in Southern African Savanna Rangelands." in: *Sustainability of Southern African Ecosystems under Global Change*, Ecological Studies, vol 248 (2024). https://doi.org/10.1007/978-3-031-10948-5_15

2020 [1] **T. Smith** and B Bookhagen. "Chapter 7: Remotely sensed rain and snowfall in the Himalaya", in: *Himalayan Weather and Climate and their Impact on the Environment*, Springer International Publishing (2020). https://www.springer.com/gp/book/9783030296834

Technical Reports

2014 [2] **T Smith**. *Climate Vulnerability in Asia's High Mountains: How climate change affects communities and ecosystems in Asia's water towers*. WWF (2014). Web Link

2009 [1] M Gale, J Kim, H Earle, A Clark, **T Smith**, K Peterson. *Open File Report VG09-5: Bedrock Geologic Map of Charlotte, Vermont* (Vermont Geologic Survey) (2009)

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Published Datasets and Software

2026 [15] **T Smith**. Discharge Deficits in the Central Himalaya Linked to Greening and Warming. Zenodo (2026). https://doi.org/10.5281/zenodo.21493187

[14] **T Smith**. Estimating the Resilience of Non-Stationary Systems. Zenodo (2026). https://doi.org/10.5281/zenodo.19731233

[13] T Liu, A Morr, S Bathiany, L Blaschke, Z Qian, C Diao, **T Smith**, & N Boers. Data gaps and outliers distort critical slowing down-based resilience indicators. Zenodo (2026). https://doi.org/10.5281/zenodo.18492399

2025 [12] **T Smith**. Predicting Instabilities in Transient Landforms and Interconnected Ecosystems. Zenodo (2025). https://doi.org/10.5281/zenodo.18031340

[11] **T Smith** and B Bookhagen. Surface-Water Temperature Trends over High Mountain Asia [Data set]. Zenodo (2025). https://doi.org/10.5281/zenodo.17630152

2023 [10] **T Smith** and N Boers. Reliability of Vegetation Resilience Estimates Depends on Biomass Density. Zenodo (2023). https://doi.org/10.5281/zenodo.7550255

[9] **T Smith** and N Boers. Global Vegetation Resilience Linked to Water Availability and Variability. Zenodo (2023). https://doi.org/10.5281/zenodo.7436669

2022 [8] **T Smith** and N Boers. Reliability of Resilience Estimation based on Multi-Instrument Time Series. Zenodo (2022). https://doi.org/10.5281/zenodo.7009414

[7] **T Smith**, D Traxl, and N Boers. Empirical evidence for recent global shifts in vegetation resilience. Zenodo (2022). https://doi.org/10.5281/zenodo.5816934

2021 [6] **T Smith** and B Bookhagen. Elevation-Snow Clusters for Glaciers and Watersheds in the Upper Indus Basin Region. Zenodo (2021). https://doi.org/10.5281/zenodo.4469473

2020 [5] **T Smith** and B Bookhagen. Global Climatic, Biotic, and Topographic Asymmetries. Zenodo (2020). https://doi.org/10.5281/zenodo.4019109

[4] **T Smith**. Topographic Asymmetry. Zenodo (2020). https://doi.org/10.5281/zenodo.3839251

[3] **T Smith** and B Bookhagen. Snow Variables for High Mountain Asia. Zenodo (2020). http://doi.org/10.5281/zenodo.3898517

2019 [2] **T Smith**, A Rheinwalt, and B Bookhagen. TopoMetricUncertainty - Calculating Topographic Metric Uncertainty and Optimal Grid Resolution. GFZ Data Services (2019). https://doi.org/10.5880/fidgeo.2019.017

2017 [1] **T Smith** and B Bookhagen. Snowmelt Parameters, 1987-2016, High Mountain Asia. GFZ Data Services (2017). https://doi.org/10.5880/fidgeo.2017.006

Invited Lectures

Dec 2026 **WMO Tipping Points Discussion Series**. *Moving Targets: Dealing with Irregular, Seasonal, and Noisy Data*

Nov 2026 **Potsdam Water Talks**. *Snow, Rivers, and Climate Change in High Mountain Asia*

Sep 2026 **BOKU, Vienna**. *Ecosystems under Pressure: Understanding Resilience in a Changing World*

Apr 2026 **ESA SIRENE Project Seminar**. *Beyond Deseasoning: New Approaches to Handling Seasonality in Dynamical Systems*

Nov 2025 **Freie Universität Berlin**. *Predicting Instabilities in Transient Landforms and Interconnected Ecosystems*

Feb 2025 **INSTAAR, CU Boulder**. *The State and Fate of Global Ecosystems: Using Dense Environmental Records to Quantify the Impacts of Climate Change*

Sep 2024 **Middlebury College**. *Snow, Rivers, and Climate Change in High Mountain Asia*

Nov 2023 **GFZ Potsdam**. *Quantifying the Influence of Snowmelt on High Alpine Rivers*

Nov 2023 **ESA EEBiomass Seminar**. *Global Shifts in Vegetation Resilience, 1980s-2010s*

Jan 2023 **Technical University of Munich**. *Biases and Pitfalls for Resilience Estimation with Satellite Vegetation Data*

Jan 2023 **Fudan University**. *Recent Snow-Water Storage Changes in High Mountain Asia*

Oct 2022 **Technical University of Munich**. *Quantifying Ecosystem Resilience with Satellite Data*

Apr 2022 **Horizon TiPES Seminar**. *Tracking Changes in Global Vegetation Resilience from Space*

Feb 2022 **Tsinghua University**. *Snow-Water Storage Changes in High Mountain Asia*

Mar 2021 **GFZ Potsdam**. *Climatic and Biotic Controls on Topographic Asymmetry at the Global Scale*

May 2018 **Humboldt University**. *Decadal Changes in the Snow Regime of High Mountain Asia, 1987-2016*

Journal Referee (selected)

Nature, Nature Climate Change, Nature Ecology & Evolution, Nature Communications, Nature Sustainability, Earth and Planetary Science Letters, Geophysical Research Letters, Science Advances, Global Change Biology, Remote Sensing of Environment, Remote Sensing, Science of the Total Environment

Advising

2024–2026 [14] **Yulong Yang** (PhD, Fudan University): Snowmelt Variations in High Mountain Asia: Characteristics, Mechanisms, and Climate Effects

2026 [13] **Nell Wassermann** (MSc Thesis): Estimation of Ice-Dammed Lake Drainage Volumes Using Surface Water and Ocean Topography (SWOT) Data

[12] **Anton Schulte-Fischedick** (MSc Thesis): Resilience Change in Southern American Tropical Dry Forests

2024 [11] **Stefan Schütz** (MSc Thesis): Satellite-Based Lake Area Time Series Analysis on the Central Andean Plateau

[10] **Juan Sebastian Valencia Velasquez** (MSc Thesis): Landslide detection using time series of spaceborne Digital Elevation Models and Lidar instruments in the South American Andes

[9] **Stepan Svintsov** (MSc Thesis): A systematic assessment of European urban agriculture potential using geoinformation

2023 [8] **Gittu Thampi** (MSc Thesis): Identify the animal pathways using a convolutional neural network

[7] **Larissa Cristina De Rezende Magalhaes** (MSc Thesis): Use of spaceborne lidar and radar data to analyze the vegetation cover in the Namibian Savannah

2022 [6] **Adeola Olanrewaju Obidiya** (MSc Thesis): Land-cover Classification and Time Series Analysis of Sentinel-1 and Sentinel-2 Images at Two Contrasting Sites (Northern Treeline vs. Taiga) with Integration of Additional Ground and Satellite Validation Data

2021 [5] **Farid Atmani** (MSc Thesis): Canopy height estimation of the Namibian Savanna Forest with ICESat-2 and GEDI missions

[4] **Toni Schmidt** (MSc Thesis): Theoretical Potential of Former, Present, and Hypothetical Optical Spaceborne Sensors for the Differentiation of Plastics Using Hyperspectral Analysis

[3] **Kittipon Wutthimetheekul** (MSc Thesis): Detection and analysis of flooding areas by using Sentinal-1 data in a part of the lower Chao-phraya river basin (Thailand)

2020 [2] **Ariane Müting** (MSc Thesis): Generating high-resolution DEMs from tri-stereo satellite imagery: A geomorphologic case study in the Quebrada del Toro, NW Argentina

2018 [1] **Ariane Müting** (BSc Thesis): Classification and spectral unmixing of remote sensing data for the Batura Glacier, Karakoram

Teaching Experience

Workshop Lead

2026 Combining Satellite and In-Situ Data for Ecosystem Monitoring, Kathmandu Link

2024 Climate Data Collection and Analysis, Kathmandu Link

Instructor, Universität Potsdam

2019–2023 Big Data Analytics (DAP03, Summer)

2019–2022 Remote Sensing of the Environment (RCM01, Winter)

2021 Spatial Data Analysis with Numerical Methods (DAP04, Winter)

2018 Terrestrial and Airborne Lidar and Photogrammetry (RSM02, Summer)

Graduate Teaching Assistant, Universität Potsdam

2017–2018 Remote Sensing of the Environment (RCM01, Winter)

2017 GIS Methods and Techniques (RCM04, Summer)

Graduate Teaching Assistant, UC Santa Barbara

2012–2014 Quantitative Geomorphology I (GEOG 237) & II (GEOG 288)

Groundwater (GEOG 116)

Water Quality (GEOG 162)

Oceans and Atmospheres (GEOG 3A)

Undergraduate Teaching Assistant, Middlebury College

2009–2011 Remote Sensing in Geoscience (GEOL 222)

The Dynamic Earth (GEOL 170)

The Ocean Floor (GEOL 142)

Theses

2018 **Smith, T** – Decadal changes in the snow regime of High Mountain Asia, 1987-2016. *Doctoral Thesis (summa cum laude), Universität Potsdam, 142. pp.* (Advisor: Bodo Bookhagen) Web Link

2014 **Smith, T** – Glacial Response to Climate Change in the Tien Shan Mountain Range of Central Asia. *Masters Thesis, UC Santa Barbara, 116. pp.* (Advisor: Bodo Bookhagen) Web Link

2011 **Smith, T** – Petrogenesis of Highly Evolved Rocks in the Springerville Volcanic Field, Eastern Arizona. *Bachelor Thesis, Middlebury College, 98. pp.* (Advisors: Ray Coish and Chris Condit)

Conference Participation

2026 **European Geophysical Union**
- **T Smith**, A Morr, B Bookhagen, and N Boers. *Predicting Instabilities in Transient Landforms and Interconnected Ecosystems*
- T Lui, **T Smith**, et al. *The influence of data gaps and outliers on resilience indicators*
- C Diao, **T Smith**, et al. *Vegetation resilience is linked to moisture availability, temperature, biodiversity and canopy complexity*
- A Schulte-Fischedick, **T Smith**, et al. *Satellite-based Tracking of Resilience Change in South American Tropical Dry Forests*

2025 **International Mountain Conference**
- **T Smith** and B Bookhagen. *Strongly Heterogeneous Surface-Water Warming Trends in High Mountain Asia*

2025 **European Geophysical Union**
- **T Smith** and B Bookhagen. *Strongly Heterogeneous Surface-Water Warming Trends in High Mountain Asia*
- Y Yang, Q You, **T Smith**. *Dynamic Impacts of Eurasian Spring Snowmelt on Summer Heat Extremes in Northern East Asia*

2024 **European Geophysical Union**
- **T Smith** and B Bookhagen. *River Temperature Gradients from Foreland to High-Mountain Environments*

2023 **European Geophysical Union**
- **T Smith** and N Boers. *How Low Can You Go? Implications of Spatial Aggregation for the Estimation of Ecosystem Resilience*

2023 **Nonlinear Data Analysis and Modeling: Advances, Applications, Perspectives**
- **T Smith** and N Boers. *Progress on Data Reliability and Processing Best Practices for Resilience Estimation with Satellite Data*

2022 **ForestSAT**
- **T Smith** and N Boers. *Global-scale Changes in Vegetation Resilience Mapped with Satellite Data*

2022 **European Geophysical Union**
- **T Smith**, N Boers, and D Traxl. *Global-scale Changes in Vegetation Resilience Mapped with Satellite Data*

2021 **European Geophysical Union**
- **T Smith** and B Bookhagen. *Climatic and Biotic Controls on Topographic Asymmetry at the Global Scale*

2021 **AI for Climate Hackathon**
- Session Organizer: *Changing Cryosphere*

2020 **European Geophysical Union**
- **T Smith** and B Bookhagen. *Shaping Planetary Surfaces: The Impact of Water Mobility on Topography*

2019 **YES Conference, Potsdam**
- Session Co-organizer – *Data-driven Remote Sensing of Earth Surface Processes*

2019 **European Geophysical Union**
- **T Smith**, A Rheinwalt, and B Bookhagen. *Determining the Optimal Grid Resolution for Topographic Analysis on an Airborne Lidar Dataset*

2018 **European Geophysical Union**
- **T Smith** and B Bookhagen. *The impacts of physical weathering regimes on large-scale slope distributions in High Mountain Asia and the Central Andes*
- **T Smith**, B Bookhagen, and A Rheinwalt. *Spatiotemporal Trends in Snow-Water Storage and the Timing of Snowmelt in High Mountain Asia*

2018 **Natural Hazards and Risks**
- **T Smith** and B Bookhagen. *Decadal trends in the timing of the snowmelt season in High Mountain Asia*

2017 **European Geophysical Union**
- **T Smith** and B Bookhagen. *Spatiotemporal Trends in the Timing and Volume of Snowfall in High Mountain Asia*

2017 **FOSDEM Open Source Conference**

2016 **European Geophysical Union**
- I Crisologo, B Bookagen, **T Smith** and M Heistermann. *Using TRMM and GPM precipitation radar for calibration of weather radars in the Philippines*

2016 **PyData, Berlin**

2015 **American Geophysical Union**

- **T Smith** and B Bookhagen. *Tracking Snowmelt Events in Remote High Asia Using Passive Microwave Data.*

2013 **American Geophysical Union**

- **T Smith** and B Bookhagen. *Glacial Retreat and Associated Glacial Lake Hazards in the High Tien Shan.*

2010 **Geological Society of America**

- **T Smith**, M Mnich, and C Condit. *Progress Towards Completed Mapping of the Springerville Volcanic Field, East-Central Arizona.*

2010 **New England Geological Society of America**

- A Clark, **T Smith**, P Ryan, J Kim, H Mango. *Elevated Arsenic in Domestic Wells from the Taconic Allochthons in Southern Vermont*
- G Springston, M Gale, J Kim, S Wright, L Becker, A Clark, **T Smith**. *Geologic Framework for Evaluating Groundwater Resources, Charlotte, VT*

2009 **Geological Society of America**

- P Ryan, J Kim, A Clark, **T Smith**, D Chow, C Sullivan, K Bright. *Ultramafic Source of Arsenic in a Fractured Bedrock Aquifer*